# FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

*Student Assessment Document*

| FINAL EXPLANATION: GENERAL SCIENCE — GRADE 10 \| SUB-STRAND 2.1: THE PERIODIC TABLE | |
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| **Student Assessment Document** | |
| **Student Name** | ______________________________________ |
| **Class** | ______________________________________ |
| **Date** | ______________________________________ |

## INSTRUCTIONS FOR STUDENTS

This is your Final Explanation task. Over the past 8 lessons, you have investigated why fertiliser bags on Kenyan farms carry such specific and different chemical formulae — and what the inner structure of an atom has to do with it. Now it is time to bring everything you have learned together into one well-reasoned, evidence-based explanation.

READ THE SCENARIO CAREFULLY before you begin writing:

Amina, a Grade 10 learner from Kericho, is helping her family prepare for the next planting season on their tea farm. She holds two fertiliser bags side by side — one labelled CAN (Calcium Ammonium Nitrate, $Ca(NO_3)_2{\cdot}NH_4NO_3$) and the other labelled DAP (Di-ammonium Phosphate, $(NH_4)_2HPO_4$). Both bags contain white crystalline solids that dissolve easily in water. Both contain nitrogen. Yet the formulae look completely different, and the agronomist at the Kericho Agricultural Office told Amina's mother that each fertiliser must be used at a specific growth stage or the tea bushes will not thrive. Amina wants to understand WHY — not just accept it.

YOUR TASK: Write a complete scientific explanation that answers the driving question:
"Why do the fertiliser bags on Kenyan farms carry such specific and different chemical formulae — and what does the inner structure of an atom have to do with it?"

Your explanation must include all four sections below. Use scientific vocabulary accurately. Support every claim with evidence or reasoning from the investigations, readings, simulations, or discussions you completed in class. Aim for depth over length — every sentence should earn its place.

## Section 1: The Inner Structure of the Atom — Building the Foundation

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| **Explain what atoms are made of and why their internal structure matters. In your response:**<br>– **Describe the three subatomic particles (proton, neutron, electron)** | Every atom has a nucleus at its centre containing positively charged protons and uncharged neutrons, surrounded by negatively charged electrons arranged in energy levels called shells. The atomic number — the number of protons — uniquely identifies each element. Nitrogen, for example, has atomic number 7, meaning every nitrogen atom has exactly 7 protons. Its 7 electrons are arranged as 2 in the first shell and 5 in the second shell (2, 5). Calcium, with atomic number 20, has the electron arrangement 2, 8, 8, 2. Phosphorus (atomic number 15) has the arrangement 2, 8, 5. The electrons in the outermost shell are called valence electrons, and they are the ones that participate in chemical bonding. Nitrogen has 5 valence electrons, calcium has 2, and phosphorus has 5. These valence electrons determine how many bonds an atom can form and therefore what ratio it will combine |

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| — their charges, masses, and locations inside the atom.<br>– Explain what the atomic number tells you and why it uniquely identifies each element.<br>– Describe how electrons are arranged in energy levels (shells) around the nucleus. Use specific examples from elements found in the two fertilisers — for example, nitrogen (N, atomic number 7), calcium (Ca, atomic number 20), phosphorus (P, atomic number 15), hydrogen (H, atomic number 1), and oxygen (O, atomic number 8).<br>– Explain what valence electrons are and why they — not the protons or neutrons in the nucleus — are the key to understanding chemical bonding and formula writing. | in — which is exactly why the formula of DAP looks different from CAN. The inner structure of the atom is not just interesting science; it is the hidden code behind every chemical formula on Amina's fertiliser bags. |

| Section 2: The Periodic Table — A Map of Electron Arrangements | |
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| **Explain how the periodic table organises elements and why that organisation is directly connected to electron arrangement. In your response:**<br>**– Describe what a period (row) and a group (column) represent in terms of electron structure.**<br>**– Explain the significance of Groups I, II, VII, and 0 (Noble Gases) in terms of valence electrons and chemical behaviour.** | The periodic table arranges all known elements in order of increasing atomic number, but its genius is that this arrangement also groups elements with similar electron configurations — and therefore similar chemical properties — into the same vertical columns called groups. Each horizontal row is called a period, and moving across a period means you are adding one electron at a time to the same outermost shell. Nitrogen sits in Period 2, Group V — it has 5 valence electrons. Phosphorus sits directly below nitrogen in Period 3, Group V — it also has 5 valence electrons. This is why nitrogen and phosphorus behave similarly in bonding: both need 3 more electrons to complete their outer shells, so both form three bonds. Calcium is in Period 4, Group II, with 2 valence electrons; it readily loses both to become $Ca^{2+}$. This pattern — elements in the same group sharing the same number of valence electrons — is called periodicity, and it is the reason the periodic table is such a powerful predictive tool. When an agronomist in Kericho knows that phosphorus is in Group V, they already know a great deal about how it will bond and what formulae its compounds will take, without needing to memorise each compound separately. |

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| **– Locate nitrogen, calcium, hydrogen, oxygen, and phosphorus on the periodic table and explain what their position tells you about their electron arrangements and likely bonding behaviour.**<br>**– Explain the concept of periodicity — why elements in the same group behave similarly — and connect this to why Kenyan farmers can sometimes substitute one fertiliser for another within the same nutrient group.** | |

| Section 3: Valence Electrons, Bonding, and the Origin of Chemical Formulae | |
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| **Use your understanding of valence electrons to explain WHY specific chemical formulae arise. In your response:**<br>**– Explain the octet rule (or duplet rule for hydrogen) and how it drives atoms to bond in fixed ratios.**<br>**– Use dot-and-cross diagrams or valence electron reasoning to show why the ammonium ion ($NH_4^+$) has the formula it does.**<br>**– Explain ionic bonding using calcium ($Ca^{2+}$) and the nitrate ion ($NO_3^-$) to show why the formula $Ca(NO_3)_2$ requires TWO nitrate ions per calcium ion.**<br>**– Explain why $(NH_4)_2HPO_4$ requires TWO ammonium ions — linking this to the charge on the hydrogen** | Atoms bond in order to achieve a full outer electron shell — 8 electrons for most elements (the octet rule) or 2 for hydrogen (the duplet rule). This drive for stability determines the exact ratio in which atoms combine, giving every compound a fixed, predictable formula. Consider the ammonium ion, $NH_4^+$: nitrogen has 5 valence electrons and needs 3 more to complete its octet, while each hydrogen atom needs just 1 more electron to complete its duplet. Nitrogen therefore bonds with 4 hydrogen atoms (one bond shares electrons with each H), using a coordinate bond for the fourth, resulting in $NH_4^+$ — a positively charged ion with a 1+ charge. Now consider $Ca(NO_3)_2$ in CAN fertiliser: calcium is in Group II and loses its 2 valence electrons to form $Ca^{2+}$ (charge: 2+). The nitrate ion $NO_3^-$ carries a charge of 1−. To balance the 2+ charge of calcium, you need exactly TWO nitrate ions — giving the formula $Ca(NO_3)_2$. In DAP, the hydrogen phosphate ion $HPO_4^{2-}$ carries a 2− charge, so it requires exactly TWO ammonium ions $NH_4^+$ (each 1+) to balance, giving $(NH_4)_2HPO_4$. Every subscript in Amina's fertiliser formulae is therefore a direct consequence of valence electron counts and ion charges — the inner structure of each atom writing the formula from the inside out. |

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| **phosphate ion ($HPO_4^{2-}$).**<br>– **Conclude by explaining how this answers the core of Amina's question: the formula is not arbitrary — it is determined by the electron structure of each atom involved.** | |

| Section 4: Answering the Driving Question — Connecting Atomic Structure to the Farm | |
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| **Now bring everything together to fully answer the driving question. In your response:**<br>– **Explain clearly why CAN and DAP have different formulae even though both contain nitrogen — referring specifically to the different elements and ion charges involved in each compound.**<br>– **Explain why these different formulae result in different nutrient availability for the tea plants (you may reference the role of nitrogen vs. phosphorus in plant growth at different stages).**<br>– **Reflect on the power of the periodic table as a tool: how does knowing an element's position on the periodic table allow a scientist — or even a well-informed farmer — to predict the formula of a compound before ever making it in a lab?**<br>– **Address Amina's original puzzle directly: the 'hidden pattern' she sensed in the formulae is real — describe what that pattern is and** | CAN and DAP both contain nitrogen, but their formulae are completely different because they contain different combinations of ions, each shaped by the electron structures of the atoms involved. In CAN, calcium ($Ca^{2+}$, Group II) and the nitrate ion ($NO_3^-$) combine in a 1:2 ratio because of their charges — a ratio fixed by electron loss and gain. In DAP, it is the ammonium ion ($NH_4^+$) and the hydrogen phosphate ion ($HPO_4^{2-}$) that combine in a 2:1 ratio, again dictated entirely by ion charges that trace back to valence electrons. Because the two fertilisers contain different ions at different concentrations, the nutrients they release into Kericho's red volcanic soil are different: CAN delivers calcium and nitrate nitrogen ($NO_3^-$), which supports leafy shoot growth in established tea bushes, while DAP delivers ammonium nitrogen ($NH_4^+$) and phosphate, which stimulates root development and is therefore applied at the early transplanting stage. The periodic table makes all of this predictable: because nitrogen and phosphorus are both in Group V, a chemist knows both will form compounds with similar structural patterns — yet the specific elements they pair with, and the charges of the resulting ions, create distinct formulae with distinct agricultural effects. The hidden pattern Amina sensed is real: it is the periodic law — the idea that an element's chemical behaviour, and therefore the formulae of its compounds, is determined by its electron arrangement, which repeats in a predictable pattern across every period of the table. The formulae on her fertiliser bags are not random labels; they are the periodic table's fingerprints, written in the language of valence electrons. |

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| **where it comes from.** | |

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| **Criterion** | **Excellent (4)** | **Proficient (3)** | **Developing (1–2)** |
| Subatomic Structure & Electron Arrangement | Accurately describes all three subatomic particles with correct charges, masses, and locations. Correctly states and explains electron arrangements for at least 4 elements from the fertilisers (e.g., N: 2,5; Ca: 2,8,8,2; P: 2,8,5; H: 1; O: 2,6). Clearly defines valence electrons and explains their role in bonding with precision. | Correctly describes subatomic particles and gives accurate electron arrangements for at least 2–3 elements. Defines valence electrons and links them to bonding with minor gaps in reasoning. | Shows partial understanding of subatomic particles; may confuse location or charge of one particle. Gives electron arrangements with errors or for fewer than 2 elements. Mentions valence electrons but does not clearly connect them to bonding or formulae. |
| Periodic Table Organisation & Periodicity | Accurately explains what periods and groups represent in terms of electron structure. Correctly identifies the periodic table positions of N, Ca, P, H, and O and links each position to valence electron count and bonding behaviour. Clearly explains periodicity and applies it meaningfully to the fertiliser context or a Kenyan agricultural example. | Correctly explains periods and groups and identifies at least 3 of the 5 elements by position. Explains periodicity with a relevant example, though the connection to the fertiliser context may be general rather than specific. | Shows basic awareness of the periodic table's structure but conflates periods and groups, or gives incorrect positions for most elements. Mentions periodicity without explaining what repeats or why. |
| Valence Electrons, Ion Charges & Formula Construction | Accurately applies the octet/duplet rule to explain bonding. Correctly derives the charges of $Ca^{2+}$, $NO_3^-$, $NH_4^+$, and $HPO_4^{2-}$ and uses charge-balancing to explain both $Ca(NO_3)_2$ and $(NH_4)_2HPO_4$ with clear, logical reasoning. Every subscript in the formulae is explicitly explained. | Correctly applies charge-balancing to explain at least one of the two formulae fully. Identifies most ion charges correctly; reasoning is mostly sound with one or two gaps. The link between valence electrons and formula construction is present but may lack full detail for one ion. | Attempts to explain formula construction but relies on memorisation rather than electron reasoning. Ion charges are stated without derivation, or charge-balancing is applied incorrectly to at least one formula. The octet rule is mentioned but not meaningfully applied. |
| Driving Question — Integration & Scientific Argument | Provides a complete, coherent answer to the driving question that integrates atomic structure, the periodic table, and ion charge-balancing into one unified explanation. Explicitly addresses why CAN and DAP have different formulae and different effects on tea plants. Identifies and names the 'hidden pattern' (periodic law / valence electron periodicity) that Amina sensed. Argument flows logically from evidence to claim | Answers most parts of the driving question with sound scientific reasoning. Connects atomic structure to formula differences and mentions the different roles of CAN and DAP in plant growth. The 'hidden pattern' is described but may not be named or fully articulated. Minor gaps in the logical chain do not undermine the overall argument. | Attempts to answer the driving question but the explanation is fragmented or relies on isolated facts rather than integrated reasoning. The connection between atomic structure and the specific formulae on the fertiliser bags is asserted rather than explained. The 'hidden pattern' is not identified or is described incorrectly. |

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| | throughout. | | |
| Use of Scientific Language & Kenya-Relevant Context | Uses all key scientific terms accurately and consistently: atomic number, electron arrangement, valence electrons, octet rule, ionic bonding, ion charge, periodicity, period, group. Grounds the explanation in the Kericho tea farm scenario throughout — not just in the introduction — and uses at least one specific local detail (e.g., soil type, growth stage, agronomist recommendation) to add authenticity and relevance. | Uses most key scientific terms correctly with only minor inaccuracies. References the fertiliser scenario in most sections. Local context is present but may feel added on rather than woven into the scientific reasoning. | Uses some scientific terms but with frequent inaccuracies or inconsistencies (e.g., confusing 'atomic number' with 'mass number', or 'valence' with 'valency' without definition). The Kericho scenario is mentioned only in the introduction or not at all. Explanation could apply to any generic chemistry lesson rather than to the specific Kenyan agricultural context. |